SamSeesSheep: A Measurement Pipeline for Sheep Ear-Angle Extraction from Ambient Pasture Video Using Foundation-Model Annotation

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Abstract

The Sheep Pain Facial Expression Scale (SPFES) [1, 2] scores five facial action units, including ear position, from photographs by trained human observers; applied over time to video, this requires a scoring decision per animal per sampled moment, a bottleneck that limits the scale and reproducibility of ovine welfare research. No automated system currently exists to extract the relevant ear-angle measurements from ambient pasture video. I present SamSeesSheep, a measurement pipeline that uses a foundation video segmentation model (SAM 3 Video) as an annotation tool, a human-in-the-loop review interface, and a lightweight edge-runnable keypoint detector (YOLO-pose, 2.5 M parameters, ∼6 MB) trained on the reviewed output. The pipeline produces per-keypoint residual standard-deviation benchmarks on genuinely held-out clips. On a clip never seen by the labeler or trainer, v0.4 achieves $\sigma = 4.06°$ (left ear) and 4.09° (right ear) for ear angle. On a second, independently selected held-out clip with a different flock arrangement and lighting, v0.7 reaches $\sigma_{\text{avg}} = 2.84°$ (left 2.39°, right 3.29°); the difference between the two clips reflects clip characteristics rather than a model improvement (on matched clips, v0.7 and v0.4 are statistically indistinguishable). Both ear-angle metrics are derived from five head keypoints (nose, left/right ear bases, left/right ear tips). Stock YOLO (`yolo26n.pt`) produces zero keypoints on both clips. The training progression from 98 to 405 reviewed instances across 3 to 8 training videos shows monotonically improving keypoint stability; beyond that, through 523 instances and 11 videos (v0.7), stability plateaus at the clip noise floor, with a documented per-keypoint regression at v0.6. I explicitly do not claim pain detection or welfare

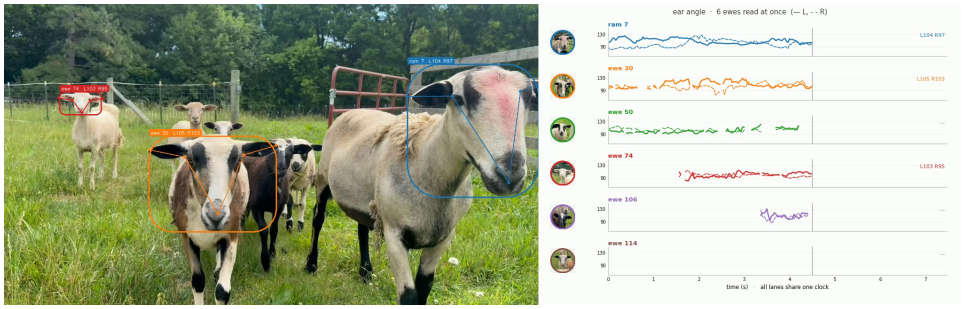


Figure 1: SamSeesSheep pipeline output on held-out clip HO-2 (Table 3): simultaneous ear-angle monitoring of six animals. *Left:* the supervision-annotated flock with bounding boxes, per-animal labels (including left/right ear angles in degrees), and five-keypoint skeletons (nose, ear bases, ear tips). *Right:* a six-lane synchronized ear-angle chart with dual traces per lane (solid = left ear, dashed = right ear) and per-animal thumbnails. The frame is a still extracted from the rendered video `synced-lanes-6ewes-pro-Test_Clip_Morning.mp4` (2036×640 px, 7.5 s; see Data Availability). The figure demonstrates the pipeline's end-to-end path: ambient pasture video → foundation-model annotation → human review → multi-animal time-series ear-angle extraction, running on an edge GPU with a ∼6 MB model.

scoring; the pipeline is a measurement instrument designed to lower the data-collection barrier for researchers who need reproducible ear-angle features from ambient footage.

1 Introduction

Sheep welfare assessment has advanced through the development of facial expression scales. The Sheep Pain Facial Expression Scale (SPFES), developed and validated by McLennan et al. [1], scores five facial areas, including ear position, that separate sheep with footrot or mastitis from healthy controls. Ear posture in particular has been established as an indicator of emotional valence in sheep by multiple independent groups [3, 4]. SPFES was validated on still photographs (one frontal and one profile view per animal per time point) scored by trained observers; it specifies no procedure for continuous video. Applying it over time at even a modest sampling rate would require a scoring decision per animal per sampled moment. Scaling

to continuous monitoring across multiple animals and days is operationally infeasible without automation.

The gap is not in the biological validity of ear-angle features; it is in the measurement infrastructure. No automated system extracts ear-angle measurements from ambient pasture video with a quantified, reproducible stability benchmark. Building one requires solving a sequence of practical computer-vision problems: detecting sheep heads in unconstrained outdoor conditions, localizing five anatomical keypoints per head (nose, left/right ear bases, left/right ear tips), tracking them across frames, and deriving a stable ear-angle signal from the keypoint geometry. Off-the-shelf object detectors provide sheep bounding boxes on roughly a third of frames but produce no landmark keypoints; the ear-angle signal is literally unmeasurable without a custom-trained keypoint head.

Foundation models have changed the economics of annotation. SAM 3 Video [5], prompted with plain-English phrases ("sheep head," "sheep ear," "sheep nose"), can segment every sheep in a clip and track them across frames. This converts the labeling task from creation (drawing boxes and placing landmarks from scratch on every frame) to review (confirming or correcting machine-generated candidates). A domain expert who knows what a sheep's ear looks like, but who is not a machine-learning engineer, can produce a reviewed keypoint dataset in hours rather than weeks.

I exploit this transformation to build a measurement pipeline: SAM 3 Video auto-segments sheep in short phone-captured clips; a labeling UI presents machine-generated keypoint candidates for human review; reviewed frames export to a YOLO-pose dataset; a small model (YOLO26n-pose, 2.5 M parameters) trains on a cloud GPU; and the resulting weights (∼6 MB) run inference on a single consumer GPU. That small footprint is deliberate: low-power edge deployment is the design target (a roadmap goal), and the benchmarks here use a 6 GB consumer card as a stand-in. The pipeline produces per-keypoint residual standard-deviation (σ) benchmarks on genuinely held-out clips, evaluated against a stock-YOLO baseline that produces zero keypoints.

Contributions.

1. A complete, open-source pipeline that converts ambient pasture video into a reviewed sheep-head keypoint dataset using a foundation model as the annotation engine and a human as the reviewer.

2. A trained edge-runnable keypoint detector (YOLO26n-pose, 2.5 M parameters, ∼6 MB) that places five keypoints on every detected sheep

head, achieving $\sim 4^\circ$ ear-angle σ on a held-out clip.

3. A reproducible benchmark protocol using genuinely held-out clips with per-keypoint residual σ, derived ear-angle σ, bootstrap confidence intervals, a window-selection sensitivity sweep, and a stock-YOLO baseline. The protocol is applied to two independent held-out clips (HO-1 and HO-2; Table 3) spanning different flock arrangements, lighting, and model versions.

4. A documented training progression (v0.2 $\rightarrow$ v0.7). Keypoint stability improves monotonically from 98 to 405 reviewed instances (3 to 8 videos, v0.2 $\rightarrow$ v0.4); beyond that it plateaus at the clip noise floor, and the progression includes a regression case (v0.6) that isolates per-keypoint labeling consistency from raw instance count. v0.7 reaches $\sigma_{\text{avg}} = 2.84^\circ$ on the second held-out clip; this is a noise-floor measurement on a different clip, not an improvement over v0.4 (Section 5.8.4).

5. A measurement instrument — not a pain detector, not a welfare scorer — designed to reduce the data-collection barrier for researchers who need reproducible ear-angle features from ambient pasture footage.

I am explicit about what this work is not. It is not a pain detection system. It is not a welfare assessment tool. It has not been validated against documented stress events. It runs on a single Katahdin flock on one farm in Delaware, USA, with one smartphone camera, annotated by a single non-veterinarian operator. Generalization to other breeds, flocks, geographies, or clinical contexts is future work requiring independent validation. This paper describes a measurement instrument that extracts ear-angle features with quantified stability, and the pipeline that built it.

2 Related Work

2.1 Ovine Facial Expression Scales and Ear Posture

McLennan et al. [1] developed the SPFES, which scores five facial areas — orbital tightening, cheek (masseter) tightening, ear position, lip and jaw profile, and nostril and philtrum shape — each as 0 (not present), 1 (partially present) or 2 (present), summed to a total pain score with a maximum of 10. In the validation study the scale separated footrot and mastitis cases from controls (AUC 0.81 and 0.80) with an inter-observer ICC of 0.86. The

ear unit is scored categorically: the ears "become fully rotated ventrally and caudally and the inner pinna of the ear becomes less visible," with the note that baseline ear carriage varies between breeds. SPFES specifies no angular thresholds. McLennan and Mahmoud [2] describe an automated SPFES pipeline (face detection, facial-landmark localization, action-unit classification, scoring); the action-unit classifiers of Lu et al. [6] used hand-crafted appearance features over eight manually annotated landmarks, including both ear tips and ear roots. Reefmann et al. [3] and Boissy et al. [4] independently established ear postures as indicators of emotional valence, finding that asymmetric or backward positions correlate with negative affective states. These studies provide the biological grounding for ear-position measurement but rely on trained human observers for scoring. The gap we address is the absence of an automated measurement pipeline.

2.2 Foundation Models for Agriculture

The Segment Anything Model family [5, 7, 8] has demonstrated strong zero-shot segmentation. SAM 3 Video extends this to the temporal domain, tracking objects across frames from text or click prompts. In agriculture, foundation models have been applied to crop monitoring, weed detection, and livestock identification [9]. Most such work uses foundation models for direct inference; our pipeline inverts this — SAM 3 Video is the annotation backbone that produces the labeled data training a lightweight deployment model.

2.3 Keypoint Detection and YOLO-Pose

YOLO-pose [10] extends YOLO object detection with a keypoint regression head, enabling simultaneous box detection and keypoint localization in one forward pass. The YOLO26n-pose variant (2.5 M parameters) suits edge deployment within consumer-GPU memory. Keypoint detection has been applied to animal pose estimation in laboratory and controlled-farm settings [11–13], but those tools assume manual annotation workflows or controlled capture. This work targets ambient, unconstrained pasture video with a foundation-model-assisted annotation pipeline.

2.4 Precision Livestock Farming

PLF systems [14] use sensors and computer vision for continuous monitoring, targeting metrics such as feeding behavior, locomotion, and social interaction. Commercial platforms operate at enterprise scale with proprietary

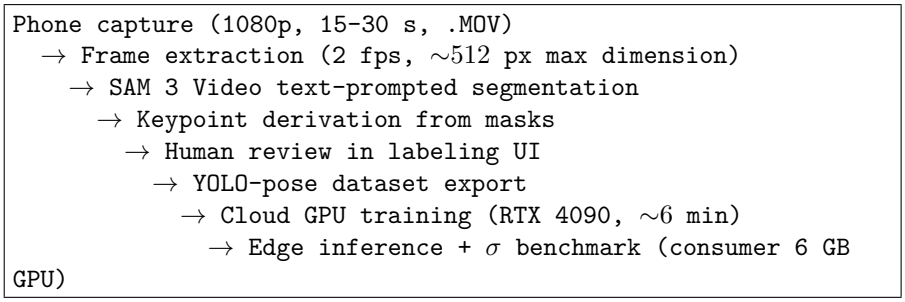


Figure 2: The five-stage SamSeesSheep pipeline. Only the trained weights cross from cloud to edge; the dataset remains on the training host.

models. This work occupies a different niche: an open-source, solo-operator-scale pipeline demonstrating the feasibility of building a domain-specific keypoint detector from scratch using foundation-model annotation, with explicit quantification of measurement stability.

2.5 The Gap

No existing system provides an end-to-end pipeline that (a) uses a foundation model for automated sheep-head keypoint annotation, (b) incorporates a human-in-the-loop review step, (c) exports a standard-format dataset for training an edge-runnable model, and (d) publishes reproducible held-out benchmarks with per-keypoint σ, derived ear-angle σ, and a stock-model baseline. This is the gap I fill.

3 Method

3.1 Pipeline Overview

The pipeline consists of five stages connected by well-defined data-format contracts, summarized in Figure 2.

3.2 Video Capture and Frame Extraction

Clips are captured at 1080p, 30 fps, typically 15–30 s. Frames are extracted at 2 fps using PyAV with a maximum dimension of 512 px (aspect-ratio preserved). The 2 fps rate and 512 px resolution keep SAM 3 Video within GPU memory budget (24 GB for the full three-session pipeline; 6 GB for the

Table 1: Keypoint slots and their derivation from SAM 3 masks.

Slot	Name	Derivation
0	Nose tip	Centroid of nose mask, or anterior point of head mask when no nose mask
1	Left ear base	Point on left ear mask closest to head-mask centroid
2	Right ear base	Point on right ear mask closest to head-mask centroid
3	Left ear tip	Point on left ear mask farthest from head-mask centroid
4	Right ear tip	Point on right ear mask farthest from head-mask centroid

reduced two-session variant) while providing sufficient temporal resolution for the sheep-head labeling task, where head pose changes slowly.

3.3 SAM 3 Video Segmentation and Keypoint Derivation

SAM 3 Video (`facebook/sam3` on Hugging Face, ∼3 GB) is called with three text prompts, each a separate propagation session: "sheep head" → head mask; "sheep ear" → left/right ear masks; "sheep nose" → nose mask. The nose session requires a 24 GB GPU; on a 6 GB GPU the pipeline drops to two sessions (head + ear) and nose keypoints fall back to head-mask geometry. SAM 3's internal tracker handles multiple instances natively. From the masks, five keypoint candidates are derived per instance per frame (Table 1). Keypoints are stored in image-space pixel coordinates; left/right is from the camera's perspective. Each keypoint carries a visibility flag v: v=0 (absent), v=1 (auto-placed, unreviewed), v=2 (human-reviewed).

3.4 Labeling UI and Review Protocol

The interface is a FastAPI backend serving a vanilla-JavaScript frontend. For each frame the reviewer sees SAM 3-derived candidates overlaid on the source image: `A` accepts all keypoints for the current instance, dragging a dot corrects placement, `Enter` saves and advances. The schema supports multi-instance frames. A single operator (the author, not a veterinarian) performed all reviews; each video was reviewed in a single session. An instance with any v=2 keypoint counts as "reviewed"; all five at v=2 is

"fully reviewed."

3.5 YOLO-Pose Export and Training

Only v=2 keypoints enter the dataset. Export produces YOLO-pose format: one `.txt` per image, one line per instance, class index 0 (`sheep_head`), normalized box, and five normalized keypoint triples (x, y, v). The `data.yaml` declares `kpt_shape: [5, 3]` and `flip_idx: [0, 2, 1, 4, 3]`, so horizontal-flip augmentation correctly swaps left/right ear keypoints. The train/val split uses deterministic MD5 hash bucketing per `(video_id, frame_idx)`: frames with bucket < 0.2 go to validation. The split is per-frame (not per-video), appropriate for a single-flock, single-camera scope where generalization to unseen poses from known scenes is the target. Training uses YOLO26n-pose (2.5 M parameters), 100 epochs, batch 8, image size 640, on an RTX 4090 via RunPod; training completes in ∼6 minutes and `best.pt` is ∼6 MB.

3.6 Edge Inference and Benchmark Protocol

For these benchmarks, inference runs on a single local consumer GPU (GTX 1660 Ti, 6 GB) — a stand-in for the low-power edge devices that are the intended deployment target, which the ∼6 MB model is sized for. The protocol, implemented in `sheep-yolo/scripts/bench_held_out.py`, evaluates keypoint stability on a genuinely held-out clip:

1. **Clip selection.** A clip with low similarity to every training video (see Data Availability for the screening procedure) and never pushed to the labeler.

2. **Target window.** A ∼5 s segment where a target sheep's head is approximately stationary; an ROI filter excludes neighbouring sheep.

3. **Residual σ.** $\sigma_{\text{residual}} = \text{std}(\text{kpt} - \text{median}_7(\text{kpt}))$. The 7-frame rolling median strips slow head drift (the sheep's actual motion) and isolates frame-to-frame jitter.

4. **Raw σ.** std(kpt) over the window, dominated by head sway; a sanity check that all models track the same physical motion.

5. **Ear-angle σ.** Ear angle is the signed angle between each ear vector (tip − base) and the head midline (nose − ear-base midpoint), via arctan2; residual σ is then taken on the detrended angle series.

Table 2: Dataset growth across versions. Validation mAP is reported from the Ultralytics training logs (see Data Availability); pixel residual σ is on the HO-1 held-out clip.

Version	Reviewed instances	Training videos	val mAP50-95 (pose)	Residual σ (mean, 5 kpts, px)
v0.2	98	3	0.479	10.89
v0.3	313	6	0.643	8.90
v0.4	405	8	0.732	7.70
v0.5	428	9	—	7.83
v0.6	471	10	—	8.09
v0.7	523	11	—	7.90

6. **Detection rate.** Any YOLO detection (confidence ≥ 0.25) within the ROI.

7. **Stock baseline.** `yolo26n.pt` (COCO-pretrained) on the same clip, reporting sheep-class detections and keypoint count (expected: zero keypoints).

4 Dataset and Experimental Setup

4.1 Dataset Characteristics

The dataset comprises video clips of a single Katahdin hair-sheep flock (5 ewes plus lambs) on a homestead pasture in Middletown, Delaware, USA. Clips were captured with one smartphone (iPhone, 1080p, 30 fps) under ambient outdoor lighting — varying sun angles, cloud cover, and shadow across sessions spanning April–May 2026. No controlled lighting, no fixed mount. This is the ambient condition the instrument is designed for.

4.2 Labeling Progression

Table 2 summarizes dataset growth. Each version is cumulative (all reviewed frames from prior versions plus new clips). The same architecture, hyperparameters, and infrastructure were used throughout; the only variables are labeled-data scale and scene diversity.

4.3 Held-Out Clips

Two clips were held out from all labeling and training: HO-1 (benchmarked v0.2–v0.7) and HO-2 (calibrated for v0.7). Both were filmed on the same pasture with the same flock but were never pushed to the labeling pod and

Table 3: Held-out clip identities. The paper IDs HO-1 and HO-2 map to the original capture filenames used throughout the released benchmark artifacts and scripts (e.g. `bench_report-IMG_3651-3way.json`).

Paper ID	Capture file	Lighting / time	Target sheep
HO-1	`IMG_3651.MOV`	Afternoon	Foreground-dominant ewe (track 34)
HO-2	`Test_Clip_Morning.mov`	Morning	Background ewe near fence (track 215)

never reviewed. Table 3 maps each paper ID to its original capture filename; the released benchmark artifacts and scripts retain those capture IDs.

HO-1 (`IMG_3651`). Filmed on the same pasture with the same flock, never pushed to the labeling pod and never reviewed. The target window spans frames 367–521 (155 frames, $\approx$ 5.17 s at 30 fps), with a single dominant foreground sheep (ByteTrack ID 34, head $\sim 234 \times 169$ px, confidence 0.6+). The ROI $(1220, 80, 1560, 390)$ is padded $\sim$half a head around the centroid range. Head-centroid standard deviation within the window is $\sim$30 px in both x and y — not perfectly still, but the calmest 5 s segment available with a large foreground head.

HO-2 (`Test_Clip_Morning`). 1920×1080, 986 frames, $\sim$30 fps, 61 MB, filmed on the same pasture with the same flock under morning lighting distinct from HO-1's afternoon light; never pushed to the pod and never reviewed. The target window spans frames 742–892 (150 frames = 5.0 s), with the target being track 215, a background tan sheep near the fence line. The ROI $(1234, 267, 1719, 538)$ isolates track 215. Head-centroid movement is $\sigma_{cx} = 47.6$ px, $\sigma_{cy} = 25.3$ px (total 53.9 px) — somewhat more motion than the HO-1 window. v0.7 detection rate within the window is 148/150 (98.7%).

4.4 Stock YOLO Baseline

`yolo26n.pt` (COCO-pretrained, identical backbone but no sheep-keypoint head) was run on both clips at confidence 0.25. It produces COCO-class `sheep` boxes on 323/933 frames (35%) for HO-1 and 118/986 frames (12%) for HO-2, but *zero keypoints* on either clip; ear angle is unmeasurable without a keypoint head. We report the COCO-`sheep`-class frame rate rather than an any-class rate: `yolo26n.pt` emits boxes across several COCO classes per clip, so an any-class rate would read higher without indicating that a sheep — let alone its ear keypoints — was localized.

Table 4: HO-1 held-out results. Stock-YOLO detection is the COCO-sheep-class rate (Section 4.4); the trained models have a single `sheep_head` class. Stock produces no keypoints.

Metric	Stock YOLO	v0.2	v0.3	v0.4
Sheep-class detection (full clip, 933 frames)	323 (35%)	909 (97%)	931 (99.8%)	933 (100%)
In-ROI detection (window, 155 frames)	0 kpts	145 (94%)	150 (97%)	149 (96%)
Residual σ (mean, 5 kpts, px)	—	10.89	8.90	7.70
Raw σ (mean, 5 kpts, px)	—	49.44	46.90	46.60

5 Results

5.1 Detection and Keypoint Stability (HO-1)

Table 4 reports the primary results on HO-1 for v0.2–v0.4 plus the stock baseline.

All three trained models detect the target on 94–97% of window frames. The detection ceiling is high from v0.2 onward because this window features a single dominant foreground sheep — an easy detection task compared with the crowded multi-sheep scenes that exposed v0.2's detection weakness in earlier benchmarks [15]. The keypoint-stability story is where the training-scale signal lives. Residual σ drops monotonically with training scale: 10.89 $\rightarrow$ 8.90 (-18%) $\rightarrow$ 7.70 px (-13% from v0.3, -29% from v0.2). Raw σ stays flat at $\sim$46–49 px, confirming that physical head sway dominates the raw measurement and that the residual metric isolates model noise.

5.2 Per-Keypoint Residual σ (HO-1)

Table 5 reports residual σ per keypoint. v0.4 improves on every keypoint. The largest gains are on ear tips — the keypoints v0.2 historically struggled to place anatomically, and the most distal from the head centroid, hence the most sensitive to mask quality and reviewer judgment. v0.4's ear-tip σ of 7.66–9.78 px on a 234-px-wide head is $\sim$3.3–4.2% of head width, down from v0.2's 5.3–5.4%.

5.3 Ear-Angle Stability (HO-1)

Table 6 reports the derived ear-angle residual σ. This is the pipeline's output scalar: a stationary sheep should produce a flat ear-angle trace, and the deviation from flatness is the measurement noise floor (Figure 3). v0.4

Table 5: Per-keypoint residual σ (px) on HO-1.

Keypoint	v0.2	v0.3	v0.4	Δ v0.2→v0.4
Nose	11.02	9.96	8.00	−27%
Left ear base	9.30	6.16	6.03	−35%
Right ear base	9.09	7.73	7.03	−23%
Left ear tip	12.63	8.41	7.66	−39%
Right ear tip	12.39	12.26	9.78	−21%
Mean	**10.89**	**8.90**	**7.70**	−29%

Table 6: Ear-angle residual σ (degrees) on HO-1.

Version	Left ear σ (°)	Right ear σ (°)
v0.2	6.71	6.07
v0.3	4.82	4.21
v0.4	4.06	4.09

achieves ∼4° jitter on both ears, a 39% (left) and 33% (right) reduction from v0.2. For reference, this project's ear-angle reference bands place "alert" at $\geq 30°$ and "down/back" at $\leq -10°$, a 40° span. These bands are an operationalization informed by the ear-posture ethograms of Reefmann et al. and Boissy et al. [3, 4]; SPFES itself scores ear position categorically and specifies no degree thresholds [1]. A 4° noise floor is ∼10% of that span — a scale comparison only, not a claim that the instrument is calibrated for SPFES scoring. The stock baseline produces zero keypoints and therefore no ear-angle measurement at all.

5.4 Training Progression and the v0.6 Regression

Subsequent versions (v0.5–v0.7) continued the labeling flywheel, all benchmarked on HO-1 (Table 7). The monotonic improvement of v0.2→v0.4 does *not* continue: from v0.4 onward both the mean residual σ and the ear-angle σ_{avg} plateau near the clip's noise floor. v0.5 reaches left 3.66° / right 4.30°. v0.6 is a mixed regression — right ear 3.55° (best recorded on HO-1) but left ear 4.65° (back to near-v0.2 levels) — showing that per-keypoint labeling consistency, not just instance count, drives stability. v0.7 recovers left to 3.70° (right 4.46°). Across all versions on HO-1, σ_{avg} improves from 6.39° (v0.2) to 4.08° (v0.4, −36%) and then oscillates within ∼0.1° of 4.0° — a

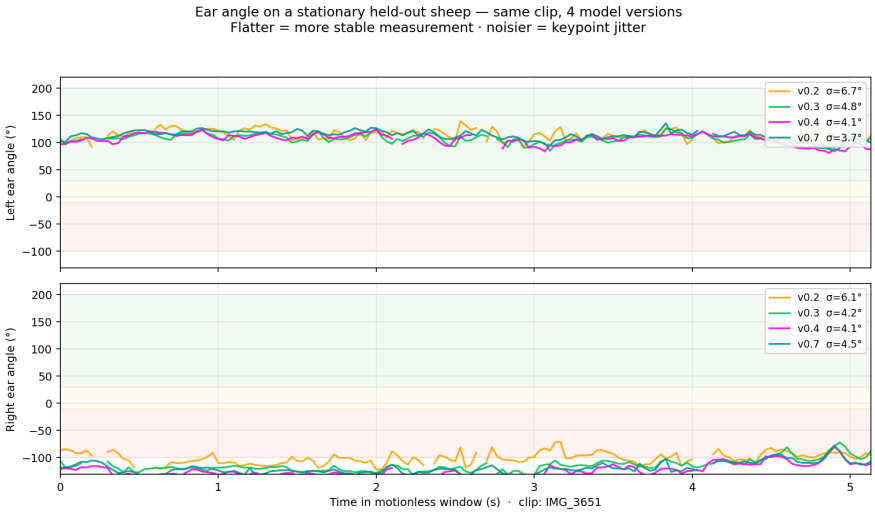


Figure 3: Ear angle over the motionless window on HO-1 for v0.2, v0.3, v0.4, and v0.7. Flatter traces indicate a more stable measurement; noisier traces indicate keypoint jitter. The large v0.2→v0.4 improvement is visible, while v0.7 overlaps v0.4 — consistent with the post-v0.4 plateau (Table 7). Legend σ values are residual standard deviations. Shaded bands are this project's ear-angle reference zones (green $\geq 30°$, amber $-10°$ to $30°$, red $\leq -10°$), informed by published ear-posture ethograms [3, 4]; they are not SPFES categories and not diagnostic cutoffs.

plateau, not continued improvement.

5.5 Statistical Robustness

Each σ above is a single point estimate from one $\sim$5 s window on one target sheep. To quantify within-window sampling uncertainty I computed moving-block bootstrap 95% confidence intervals (block = 10 frames, 5000 resamples, on the detrended residual series; Table 8). The intervals are wide and adjacent-version intervals overlap almost completely: the gross v0.2→v0.4 improvement clears the band, but the v0.4↔v0.7 differences do not. Two consequences follow, and the paper's claims are calibrated to them. First, the training-scale signal is real only at coarse resolution (roughly, 98 vs. 405 reviewed instances), not between adjacent versions. Second, these intervals capture only within-window sampling; they do *not* capture the larger structural uncertainty of a single window, single target sheep, and single clip per

Table 7: Full progression on HO-1. Improvement is monotonic only through v0.4; v0.5–v0.7 plateau, with v0.6 a per-keypoint regression.

Version	Left ear σ (°)	Right ear σ (°)	σ_{avg} (°)	Residual σ mean (px)
v0.2	6.71	6.07	6.39	10.89
v0.3	4.82	4.21	4.52	8.90
v0.4	4.06	4.09	4.07	7.70
v0.5	3.66	4.30	3.98	7.83
v0.6	4.65	3.55	4.10	8.09
v0.7	3.70	4.46	4.08	7.90

Table 8: Moving-block bootstrap 95% CIs on ear-angle residual σ (degrees). Adjacent-version intervals overlap; only the gross v0.2→v0.4 change is separable.

Clip	Version	Left ear σ (95% CI)	Right ear σ (95% CI)
HO-1	v0.2	6.71 [5.56, 8.04]	6.07 [4.65, 7.60]
HO-1	v0.4	4.06 [2.97, 4.99]	4.09 [2.83, 5.26]
HO-1	v0.7	3.70 [2.96, 4.10]	4.46 [3.08, 5.64]
HO-2	v0.2	5.37 [1.31, 8.82]	2.24 [0.95, 3.09]
HO-2	v0.4	2.50 [1.15, 3.35]	2.72 [1.22, 3.69]

condition, which no resampling of one window can estimate.

5.6 Window-Selection Sensitivity

Each headline σ is computed on one hand-picked "calmest ∼5 s" window, which raises a fair concern: selecting the window on the variable being minimized (head motion) could bias the reported noise floor downward. To bound this effect I slid the same fixed-length window across the target sheep's entire detected span (step 2 frames; windows retained when ≥ 60 frames carry a computable ear angle) and recomputed σ_{avg} and the within-window head motion (raw positional σ) for every window. The procedure is implemented in `sheep-yolo/scripts/window_sensitivity.py` and reproduces each paper window's committed σ_{avg} and raw σ exactly. Table 9 and Figure 4 report the resulting distributions.

The hand-picked window does sit low in the HO-1 v0.4 distribution (18th percentile), confirming the selection mechanism is real — but its magnitude is small. The median window scores 4.55° against the reported 4.07°, and

Table 9: Window-selection sensitivity. The same fixed-length window is slid across each target's detected span; σ_{avg} is the mean left/right ear-angle residual σ per window. "Pctile" is the fraction of windows whose σ_{avg} is at or below the hand-picked paper window's — low means the paper window is among the calmer (lower-σ) ones.

Run	Windows	Paper σ_{avg} (°)	Pctile	Median σ_{avg} (°)	Max σ_{avg} (°)	corr(motion, σ)
HO-1 v0.4	200	4.07	18th	4.55	5.69	0.56
HO-1 v0.7	205	4.08	43rd	4.43	8.43	0.47
HO-2 v0.4	324	2.61	75th	2.37	6.32	0.29

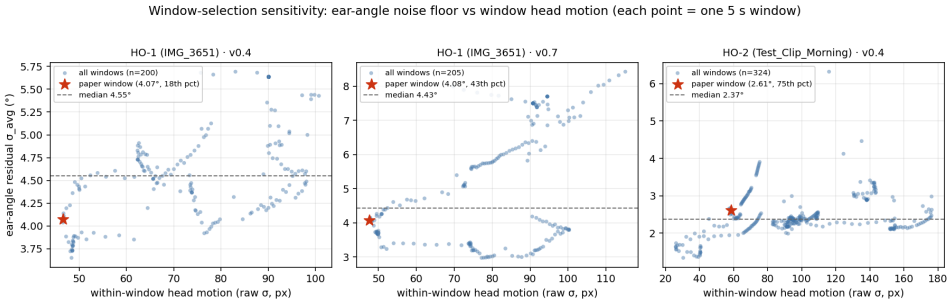


Figure 4: Ear-angle residual σ_{avg} for every 5 s window slid across the target's detected span, versus within-window head motion. Red star: the hand-picked paper window; dashed line: the median window. Across all windows and both clips σ_{avg} stays a small fraction of the 40° reference-band span and rises only modestly with head motion — the reported noise floor is not an artifact of selecting the stillest segment.

even the noisiest 5 s window of that sheep reaches only 5.69° (∼14% of the 40° reference span). For v0.7 the paper window is mid-distribution (43rd percentile), and on HO-2 it is above the median (75th percentile) — a higher-motion window than typical, so the reported 2.61° is, if anything, conservative. Within-window motion correlates with σ_{avg} only moderately ($r = 0.29$–0.56), so motion explains part but not most of the window-to-window variance. The noise floor is therefore not an artifact of selecting the stillest segment: across every window of both targets σ_{avg} stays between ∼1.3° and ∼8.4°, a small fraction of the reference-band span in all cases.

Table 10: HO-2 held-out results. Stock-YOLO detection is the COCO-sheep-class rate (Section 4.4); the any-class rate (100%) is not reported as a detection figure because it does not reflect sheep localization.

Metric	Stock YOLO	v0.2	v0.3	v0.4
Sheep-class detection (full clip, 986 frames)	118 (12%)	929 (94%)	986 (100%)	986 (100%)
In-ROI detection (window, 150 frames)	0 kpts	132 (88%)	149 (99%)	150 (100%)
Residual σ (mean, 5 kpts, px)	—	6.73	4.22	3.85
Raw σ (mean, 5 kpts, px)	—	41.53	56.09	58.35

5.7 v0.4 on Held-Out versus In-Distribution

v0.4's held-out residual σ of 7.70 px approaches v0.3's in-distribution numbers on its own training clips (4.10–5.92 px, reported in the v0.3 benchmark [15]). The $\sim$1.8–3.6 px gap between in-distribution and held-out performance is the generalization cost for an unseen scene; that this gap is quantifiable at all was unavailable from the earlier within-distribution benchmarks.

5.8 Second Held-Out Clip: HO-2

The second clip provides a cross-clip validation point, testing whether the training-scale monotonicity observed on HO-1 (v0.2→v0.4) reproduces on an independently selected clip with different flock arrangement, lighting, and target. Figure 1 shows the pipeline's multi-animal output on this clip. By design, HO-2 carries the v0.2–v0.4 trend-reproduction set plus v0.7 as the then-current model; v0.5 and v0.6 were benchmarked only on HO-1 (the full ladder lives in Table 7). The v0.7 run on HO-2 recorded the aggregate ear-angle σ (Table 12) rather than a per-keypoint breakdown, so the per-keypoint Table 11 lists v0.2–v0.4.

5.8.1 Detection and Keypoint Stability

Table 10 reports results for v0.2–v0.4 plus the stock baseline. The detection rate climbs from 88% (v0.2) to 100% (v0.4). Residual σ drops monotonically: 6.73 → 4.22 (−37%) → 3.85 px (−9% from v0.3, −43% from v0.2) — the same monotonic v0.2→v0.4 pattern as HO-1. Raw σ varies 41–58 px, driven by lateral head drift rather than model noise.

Table 11: Per-keypoint residual σ (px) on HO-2. Per-keypoint data exist for v0.2–v0.4 on HO-2; see Section 5.8 for the version coverage.

Keypoint	v0.2	v0.3	v0.4	Δ v0.2→v0.4
Nose	4.07	3.75	4.18	+3%
Left ear base	3.46	2.54	2.36	−32%
Right ear base	8.10	2.91	2.63	−68%
Left ear tip	6.21	6.12	4.38	−29%
Right ear tip	11.79	5.79	5.69	−52%
Mean	**6.73**	**4.22**	**3.85**	−43%

5.8.2 Per-Keypoint Residual σ

Table 11 reports per-keypoint residual σ. The pattern differs from HO-1: v0.2's largest variance is on the right ear (base 8.10 px, tip 11.79 px), while the left side and nose are already reasonably stable at 98 instances. This asymmetry likely reflects the target's pose and the morning lighting angle. By v0.3 the right-ear asymmetry largely resolves, and by v0.4 left/right ear-base σ are nearly identical (2.36 vs. 2.63 px), suggesting the model has learned symmetric ear placement from diverse views. v0.4's ear-tip σ of 4.38–5.69 px on a 316-px-wide head is ∼1.4–1.8% of head width, versus 3.3–4.2% on HO-1's 234-px-wide head; the larger apparent head (∼1.35×) partly but not fully explains the lower pixel σ.

5.8.3 Ear-Angle Stability

Table 12 reports ear-angle residual σ. On HO-1 the left/right σ were roughly symmetric; here v0.2 shows a pronounced asymmetry (5.37° left vs. 2.24° right) that resolves by v0.3 and re-emerges smaller in v0.4 and v0.7. This may reflect genuine anatomical asymmetry in this sheep's ear carriage or a systematic left/right bias from the viewing angle. v0.7's $\sigma_{\text{avg}} = 2.84°$ is 7.1% of the 40° reference-band span. Note that on this clip v0.4's σ_{avg} is 2.61° — slightly lower than v0.7's 2.84°; the two are within each other's bootstrap intervals (Table 8), so v0.7 is not an improvement over v0.4 here.

5.8.4 Two-Clip Cross-Comparison

Table 13 compares ear-angle σ_{avg} across the two clips at equivalent training scale. At identical scale (405 instances, 8 videos), v0.4 achieves $\sigma_{\text{avg}} = 4.08°$

Table 12: Ear-angle residual σ (degrees) on HO-2.

Version	Left ear σ (°)	Right ear σ (°)
v0.2	5.37	2.24
v0.3	2.49	2.08
v0.4	2.50	2.72
v0.7	2.39	3.29

Table 13: Cross-clip comparison of ear-angle σ_{avg}. The HO-1 vs. HO-2 gap is a clip-difficulty effect, not a model improvement.

Metric	v0.4 on HO-1	v0.4 on HO-2	v0.7 on HO-2
Window size	155 fr (5.2 s)	150 fr (5.0 s)	150 fr (5.0 s)
Target head size (approx.)	234×169 px	316×185 px	316×185 px
Target type	Foreground dominant	Background near fence	Background near fence
Head centroid σ	$\sim$30 px (x, y)	48 px (x), 25 (y)	48 px (x), 25 (y)
In-ROI detection rate	96%	100%	98.7%
Residual σ (mean, px)	7.70	3.85	—
$\sigma_{\text{left ear}}$ (°)	4.06	2.50	2.39
$\sigma_{\text{right ear}}$ (°)	4.09	2.72	3.29
$\boldsymbol{\sigma_{\text{avg}}}$ **(°)**	**4.08**	**2.61**	**2.84**
σ_{avg} as % of 40° ref. span	10.2%	6.5%	7.1%
Reviewed instances	405	405	523
Training videos	8	8	11

on HO-1 and 2.61° on HO-2 — a 36% difference attributable to the clip, not the model. Three factors likely contribute: the larger apparent head on HO-2 ($\sim$1.35$\times$ wider) reduces pixel σ for a fixed angular jitter; the target is more isolated; and the morning lighting gives higher tan-on-green contrast. This is precisely why the cross-clip number must not be read as a version improvement: **the** 2.84° **reached by v0.7 on HO-2 is a noise-floor measurement on an easier clip, and on matched clips v0.7 and v0.4 are statistically indistinguishable** (Table 8). A single held-out clip provides a lower bound rather than a point estimate of model performance; the two-clip comparison makes this explicit.

6 Discussion

6.1 What the Numbers Mean

The headline result — $\sim$4.08° ear-angle σ_{avg} on HO-1 (v0.4) and $\sim$2.84° on HO-2 (v0.7) — establishes a noise floor for automated ear-angle mea-

surement from ambient pasture video. Whether 2.8–4.1° is "good enough" depends entirely on the application. For detecting gross ear-position changes (a sustained shift from +30° alert to −10° down/back, a 40° range), 4° jitter is 10% of the decision range and unlikely to generate false positives. For subtle within-band changes (a 5° shift within the neutral zone), 4° jitter would drown the signal. The appropriate use is therefore coarse temporal patterns — sustained ear-position state changes over minutes, not frame-to-frame micro-movements. The stock baseline is instructive: a practitioner using off-the-shelf models would obtain exactly zero ear-angle data points. The keypoint head must be grown against the target animals; there is no pretrained shortcut.

6.2 Limitations

The following limitations, drawn from the project's `VALIDATION.md` contract, constrain any claim made on these results.

Single flock, single breed, single geography. The dataset is ∼5 Katahdin hair-sheep ewes (plus lambs) on one homestead in Middletown, Delaware. No claim transfers to other breeds (wool, horned, cropped or floppy ears), geographies, or management conditions without independent validation.

Single operator, single camera. One non-veterinarian operator captured all video with one smartphone and performed all reviews. There is no inter-annotator agreement, no independent ground truth, and no vet-validated keypoint placement. Labeling consistency is measured only indirectly through the training progression. Consequently the reported σ quantify *precision* (frame-to-frame self-consistency), not accuracy: with no independent ground-truth landmark, a fixed systematic error in keypoint placement would be invisible to this benchmark, as would any slow bias absorbed by the 7-frame rolling median.

Point estimates, no significance on fine deltas. Each σ is a single point estimate from one window on one sheep. As Section 5.5 shows, adjacent-version differences fall within the within-window bootstrap band; only the gross v0.2→v0.4 change is separable. No version ranking finer than that should be inferred, and the bootstrap does not address single-clip/single-window structural uncertainty.

Window selection. The benchmark window is the calmest available ∼5 s segment, which could bias the noise floor downward. The sensitivity sweep of Section 5.6 bounds this effect: it is real but small (the HO-1 v0.4 window is at the 18th percentile, ∼0.5° below the median window, and even the noisiest window of that target is 5.69°), and on HO-2 the chosen window is actually above the median. The reported floors are best-to-typical, not unrepresentative outliers.

Two held-out clips, same flock and same animals. The protocol uses two held-out clips spanning different lighting and arrangements, but both are from the same flock, pasture, and camera — and both show the same five individuals. "Held-out" therefore demonstrates generalization to unseen poses, scenes, and lighting on *known* animals, not to unseen animals; animal-level held-out evaluation requires a second flock. A multi-clip ablation (3–5 clips from different sessions and weather) would strengthen the generalization claim.

Ambient-vs-clinical gap. SPFES [1] was validated in clinical pain conditions (footrot, mastitis) with trained observers scoring still photographs, and the ear-posture ethograms behind this project's reference bands [3, 4] come from controlled settings. Generalizing them to ambient pasture observation — where sheep move their ears for vigilance, social signaling, insect avoidance, and wind — is an unresolved scientific question. This instrument can extract ear angles; it cannot tell you what they mean.

The measurement is not the diagnosis. Ear angle is one geometric feature. Welfare is a multi-dimensional construct requiring veterinary assessment [14]; pain is a clinical determination requiring multiple indicators. This pipeline measures one feature. It is not a welfare instrument.

No student-vs-teacher comparison. The trained detector is not benchmarked against the SAM 3 teacher labels it distilled from, so whether the student matches, exceeds, or trails the teacher on held-out frames is unmeasured (SAM 3 was never run on the held-out clips, by design). A student-vs-teacher agreement study on held-out frames is the most direct next ablation.

6.3 Known Failure Modes

- **Dark fleece / low contrast.** SAM 3 segmentation degrades on black-faced sheep at dusk; the trained model inherits this from its

light-fleeced-only training data.

- **Motion blur.** Handheld capture of moving sheep is the median case; heavy blur degrades segmentation and keypoint stability.
- **Extreme angles.** Best on frontal and three-quarter views; profile views and full ear occlusion reduce keypoint availability.
- **Occlusion.** In crowded frames the model may detect the wrong sheep or place keypoints on occluded anatomy.
- **Wool-covered ears.** Heavy fleece can obscure ear shape entirely; untested on wool breeds.

6.4 What Validation Against Stress Events Would Require

Turning this instrument into a welfare-relevant tool requires: documented stress events with timestamps (hoof trimming, tagging, separation, transport); continuous pre/post-event monitoring; a within-animal delta analysis; and a pre-registered kill criterion. The project's `VALIDATION.md` specifies that if fewer than 70% of documented stress events show a measurable change in ear-angle features, the welfare project terminates and the write-up of what failed becomes the deliverable. This paper describes the measurement instrument; that follow-up project would determine whether the measurements are welfare-informative.

6.5 Negative Results as Features

This work deliberately publishes negative results. The stock baseline (zero keypoints, 12–35% sheep detection) establishes that off-the-shelf models are not a viable shortcut. The v0.6 regression (right ear at all-time best 3.55° but left ear regressing to 4.65°) demonstrates that more data does not monotonically improve all keypoints; per-keypoint coverage and per-session labeling consistency matter at least as much as instance count. The post-v0.4 plateau (Section 5.4) is reported rather than hidden behind a selectively quoted best version. And the v0.3 benchmark correction — admitting that the originally-claimed "held-out" clips were actually in v0.3's training distribution — is published alongside the corrected v0.4 benchmark [15]. These are evidence that the measurement methodology is working as designed.

7 Conclusion

I have presented SamSeesSheep, a measurement pipeline that converts ambient pasture video into quantified ear-angle features using a foundation-model annotation flywheel. The pipeline produces a small edge-runnable model (2.5 M parameters, ∼6 MB) that places five keypoints on every detected sheep head and achieves ear-angle residual σ_{avg} of 4.08° (v0.4 on HO-1) and 2.84° (v0.7 on HO-2) — a metric literally unmeasurable with off-the-shelf object detectors. The training progression shows monotonic improvement in keypoint stability from 98 to 405 reviewed instances (v0.2 to v0.4), then a plateau at the clip noise floor with a documented v0.6 regression; the benchmark protocol, applied to two independent held-out clips with bootstrap confidence intervals, provides a reproducible methodology for quantifying measurement quality and its uncertainty.

I am explicit about the boundary: this is a measurement instrument, not a pain detector or welfare scorer. It measures ear-angle features with quantified stability. Whether those features are welfare-informative is a question for the follow-up validation study this instrument is designed to enable. Future work includes additional held-out clips spanning varied weather and camera positions; cross-flock generalization on wool breeds and other geographies; validation against documented stress events with the kill criterion above; inter-annotator agreement measurement; and continuous-monitoring deployment at water troughs or handling chutes, where sustained ear-position patterns become the unit of measurement.

Data and Code Availability

The author's pipeline code, benchmark scripts, machine-readable benchmark artifacts, paper source, and documentation are available under the MIT License at `https://github.com/antonemking/SamSeesSheep`; the trained YOLO-pose weights carry a separate license inherited from Ultralytics (see Companion artifacts below). Reproducibility is layered, so each headline number can be re-derived at the level a reader is equipped for.

In the repository (clone and verify). Every σ, detection rate, and per-keypoint number in this paper is stored in committed JSON artifacts under `sheep-yolo/artifacts/` and is checkable in one command with `sheep-yolo/scripts/verify_paper_claims.py`, which reads only those JSONs and re-validates each value against the paper (59/59 checks pass

on a fresh clone). The bootstrap confidence intervals of Table 8 are in `bench_bootstrap_ci.json`.

Companion artifacts (weights and cached predictions). The trained per-version weights (`sheep-pose-v0.{2-7}-yolo26n`) are git-ignored for size and published on the Hugging Face Hub at `https://huggingface.co/antking1/sheep-pose-yolo26n`, with a model card and a persistent DOI (10.57967/hf/9129) pinned to a fixed revision. Because they are trained with Ultralytics YOLO (AGPL-3.0), the weights are released under AGPL-3.0 rather than MIT; only the author's own repository code is MIT-licensed. The per-frame model-prediction caches (`sheep-yolo/artifacts/_cache/*.pkl`, ∼3 MB total) ship as an asset on the `paper-artifacts-v0.7` GitHub release (`https://github.com/antonemking/SamSeesSheep/releases/tag/paper-artifacts-v0.7`). With the caches, every committed benchmark JSON can be regenerated from scratch (`gen_bench_Test_Clip_Morning.py` for HO-2; `gen_residual_px.py` and the archived `bench_v02_v03.py` / `bench_v04_v05_v0{6,7}.py` for HO-1), and the ear-angle chart of Figure 3 with `gen_ear_angle_chart.py`. The hero still of Figure 1 is a frame from `synced-lanes-6ewes-pro-Test_Clip_Morning.mp4`, also in the release.

Available on request (raw video and dataset). The held-out clips (HO-1, HO-2) and the full reviewed-keypoint training dataset are retained off-repository for size and to limit redistribution of identifiable homestead footage; they are available from the author on request. A reader with the raw clips and the released weights can reproduce the cached predictions themselves; a reader without them reproduces every published number from the caches onward.

Reported from outside the verification harness. Three figures are sourced outside the repository artifacts and are disclosed as such: the validation mAP50-95 values (Table 2) come from the Ultralytics training logs produced on the cloud GPU (only the weights, not the logs, return to the laptop); the in-distribution 4.10–5.92 px residual-σ range (Section 5.7) is quoted from the prior v0.3 benchmark report [15], whose per-clip JSONs are training-set data held off-repository; and the held-out clips were screened by pixel normalized cross-correlation (NCC) of the first frame against every training video — HO-1 scored NCC <0.23 — and were held in a separate `test-clips/` directory, never pushed to the labeler (enforced by per-file

checksum dedup). The training-video frames needed to recompute these NCC scores are themselves off-repository, so this screening remains an author attestation the public artifacts cannot independently re-derive.

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Note on scope

This work was conducted on a single Katahdin flock in Middletown, Delaware, using one smartphone camera and one human reviewer (the author, who is not a veterinarian). All claims are bounded by the scope described in the project's `VALIDATION.md` contract. No pain detection. No welfare scoring. A measurement instrument.